

EHGIZ

PLATFORM

Software Project Document

Peer-to-Peer Rental Marketplace · AI-Enhanced

Project Type: P2P Rental Marketplace (AI-Enhanced)

01 Project Overview

Project Name	Ehgiz
Project Type	Peer-to-Peer (P2P) Rental Marketplace (AI-Enhanced)

Ehgiz is a digital platform that enables individuals to rent tools and equipment from each other instead of purchasing them. The platform integrates AI-powered agents to improve search accuracy, pricing recommendations, and overall user experience.

02 Problem Statement

- Tools and equipment are expensive and often underutilized.
- Users frequently need tools for short-term use.
- There is no structured and trusted rental platform in Egypt.

Informal renting lacks:

- Pricing standards
- Trust mechanisms
- Safety and guarantees

03 Proposed Solution

Ehgiz provides a structured platform that:

- Connects tool owners with renters
- Enables secure booking and communication

Uses AI to:

- Understand user intent
- Recommend relevant tools
- Suggest fair pricing
- Detect suspicious behavior
- Implements an escrow-based trust system

04 Target Audience

- Individuals with rarely used tools
- Students and DIY users
- Small business owners
- Freelancers (construction, photography, maintenance)

05 System Features

◆ 5.1 User Features

- User Registration & Authentication
- Tool Listing with Images and Details
- Search by Category and Location
- Booking System (Calendar-based)
- Chat Between Users
- Ratings and Reviews

◆ 5.2 Advanced Features

- Escrow Payment System
- Optional Tool Insurance
- SOS / Issue Reporting
- Rental History Tracking

06 AI Features

◆ 6.1 Classification Agent

- Classifies user input into tool categories

Example:

Input	Output
"عايز أفتح حيطه"	Drill

◆ 6.2 Pricing Agent

Suggests rental price based on:

- Market averages

- Tool condition
- Demand level

◆ 6.3 Image Analysis

- Detects tool type from uploaded image
- Evaluates condition
- Identifies visible damage

◆ 6.4 RAG Knowledge Base

Uses embeddings and vector search. Provides:

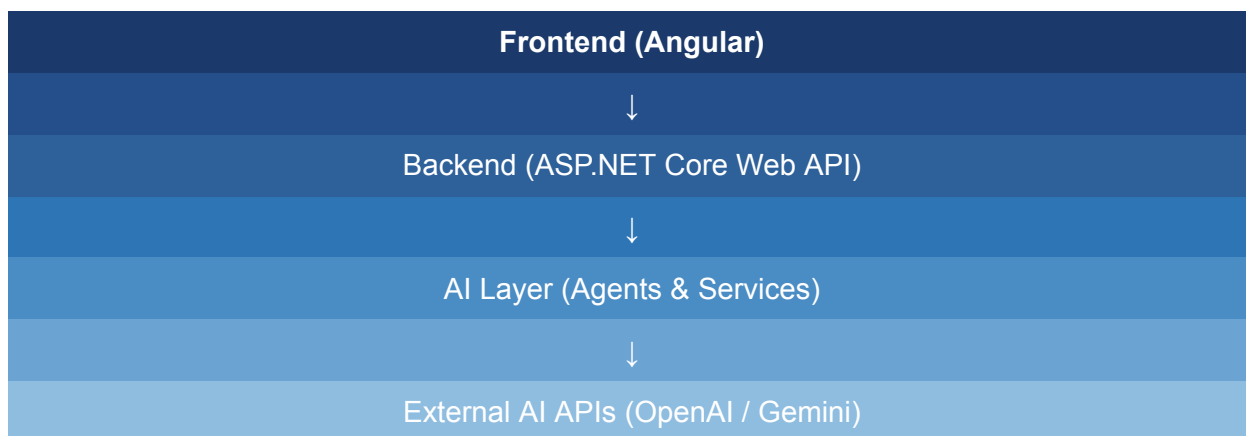
- Smart recommendations
- Tool usage guidance
- Context-aware search

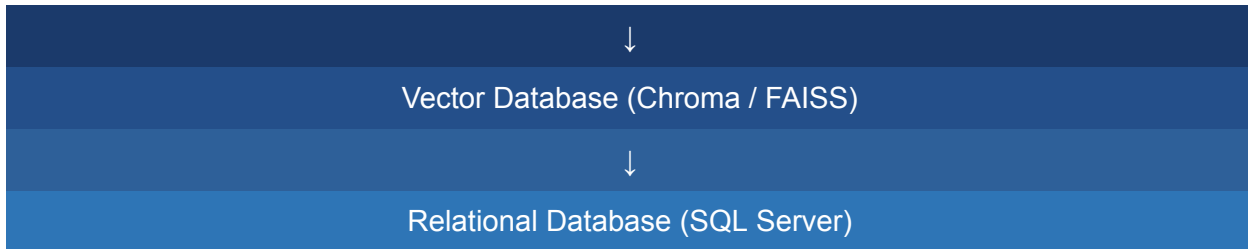
◆ 6.5 Image-to-Search

- Allows users to upload an image
- Finds similar tools available nearby

07 System Architecture

◆ 7.1 High-Level Architecture





08 Backend Architecture (N-Tier Architecture)

◆ Layers

- Presentation Layer: Controllers (API Endpoints)
- Business Layer: Services and AI Agents
- Data Access Layer: Repositories , Database
- Domain Layer: Entities , DTOS

09 AI System Design

◆ Agent Structure

Each agent consists of:

- Prompt (instruction logic)
- Service (execution layer)
- Tools (DB queries / APIs)

◆ Key Principle

- No model training required

System relies on:

- Pre-trained LLM APIs
- Embeddings for semantic search

10 Database Design

◆ Main Tables

- Users
- Tools
- Categories
- Bookings
- Reviews
- Payments
- Messages

11 Security & Trust

- JWT Authentication
- Role-Based Authorization
- Escrow Payment Handling
- Secure API Key Management

12 Technology Stack

◆ Backend

- ASP.NET Core Web API
- Entity Framework Core
- SQL Server

◆ Frontend

- Angular

◆ AI & Data

- OpenAI API (GPT-4o / Vision)
- Embeddings API
- Vector Database (Chroma / FAISS)

13 Testing Strategy

- Unit Testing for Services
- Integration Testing for APIs
- End-to-End Testing for User Flows
- Manual Testing for UI/UX

14 Future Enhancements

- Mobile Application
- Real-time Tracking
- Payment Gateway Integration
- Advanced Recommendation System

15 Conclusion

Ehgiz is a scalable and practical solution that leverages modern software architecture and AI technologies to solve a real-world problem. The platform enhances accessibility, reduces cost, and builds trust in peer-to-peer tool sharing.